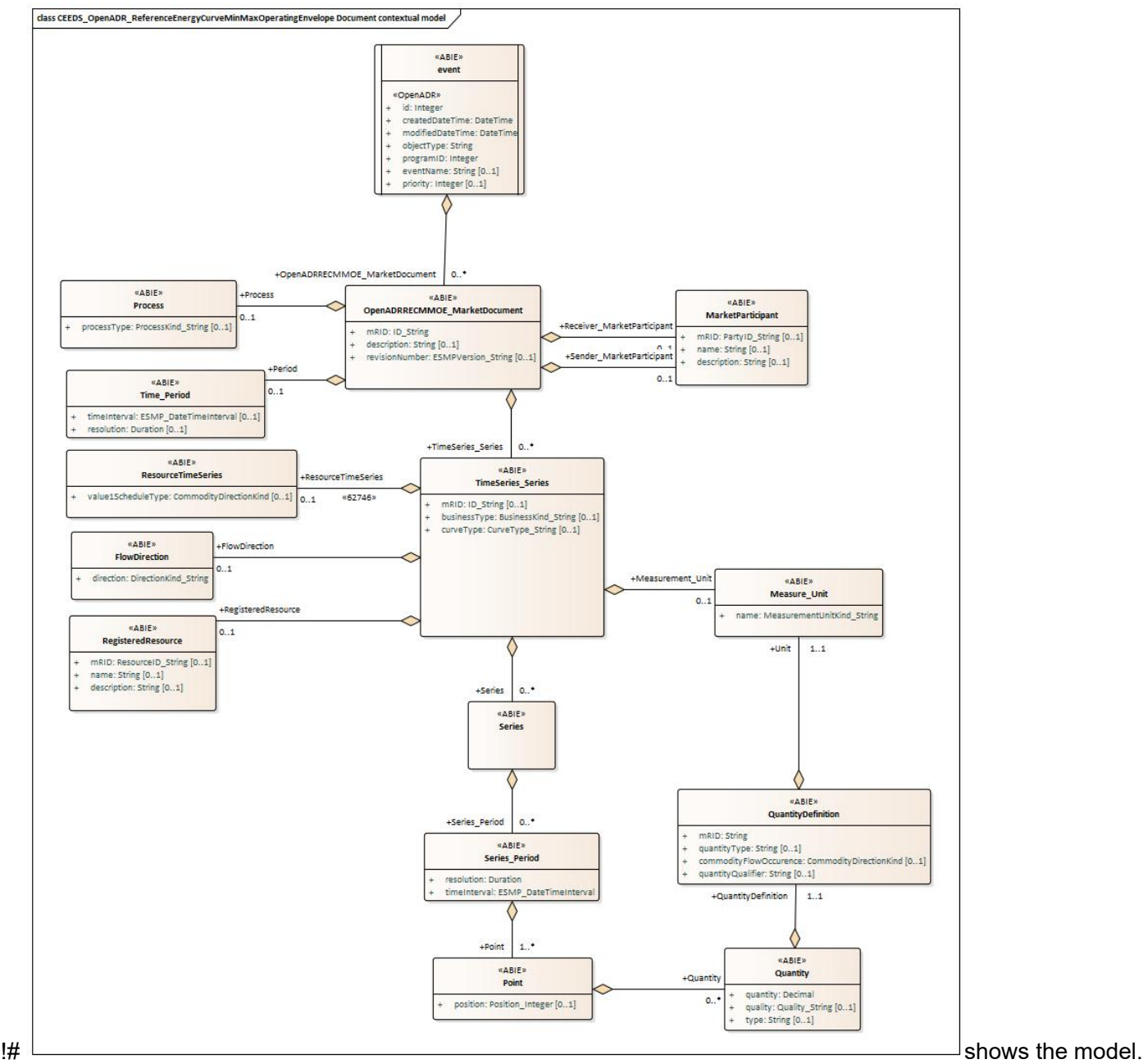


CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model

Overview of the model



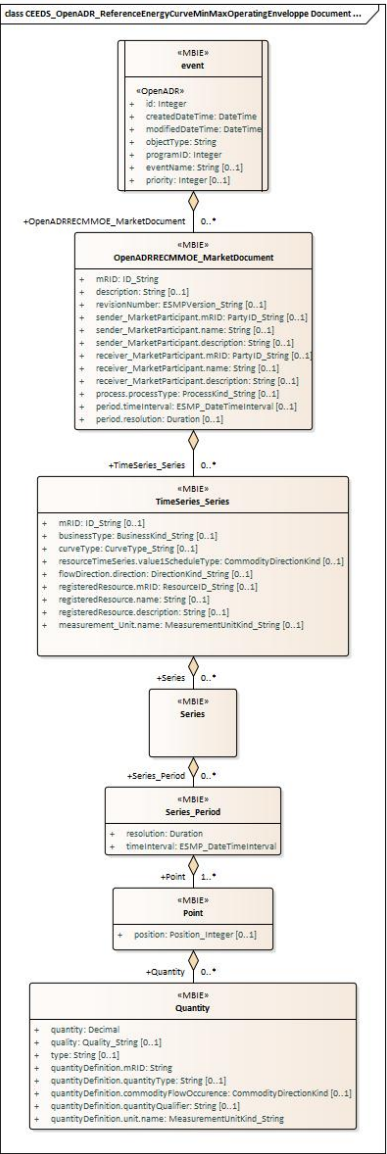


Figure !# - CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model

IsBasedOn relationships from the European style market profile

!# shows the traceability dependency of the classes used in this package towards the upper level.

Table !# - IsBasedOn dependency

Name	Complete IsBasedOn Path
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event	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::event
OpenADRRECMMOE_MarketDocument	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::OpenADRRECMMOE_MarketDocument
Point	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Point
Quantity	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Quantity
Series	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Series
Series_Period	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Series_Period
TimeSeries_Series	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::TimeSeries_Series

##MXM##

Detailed CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model

event root class

!# shows all attributes of event.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::event

Order	mult.	Attribute name / Attribute type	Description
0	[1..1]	id Integer	Identification of the event
1	[1..1]	createdDateTime DateTime	Date and time this event has been created
2	[1..1]	modifiedDateTime DateTime	Date and time this event has been modified
3	[1..1]	objectType String	Used as discriminator.
4	[1..1]	programID Integer	ID attribute of the program object the event is associated with.
5	[0..1]	eventName String	User defined string for use in debugging or User Interface.
6	[0..1]	priority Integer	Relative priority of event. A lower number is a higher priority.

##MXM##

!# shows all association ends of event with other classes.

Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::event with other classes

Order	mult.	Class name / Role	Description
7	[0..*]	OpenADRRECMMOE_MarketDocument OpenADRRECMMOE_MarketDocument	Association Based On : CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::event.[] --- CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::OpenADRRECMMOE_MarketDocument.OpenADRRECMMOE_MarketDocument[0..*]

##MXM##

OpenADRRECMMOE_MarketDocument

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of OpenADRRECMMOE_MarketDocument.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly
model::OpenADRRECMMOE_MarketDocument

Order mult. Attribute name / Attribute type		Description
0	[1..1] mRID ID_String	The unique identification of the document being exchanged within a business process flow. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such as document identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
1	[0..1] description String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
2	[0..1] revisionNumber ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
3	[0..1] sender_MarketParticipant.mRID PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
4	[0..1] sender_MarketParticipant.name String	The name is any free human readable and possibly non unique text naming the object.
5	[0..1] sender_MarketParticipant.description	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

String		
6	[0..1] receiver_MarketParticipant.mRID	The identification of a party in the energy market.
	PartyID_String	In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
7	[0..1] receiver_MarketParticipant.name	The name is any free human readable and possibly non unique text naming the object.
String		
8	[0..1] receiver_MarketParticipant.description	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
String		
9	[0..1] process.processType	The identification of the nature of process that the document addresses. --- The Process associated with an electronic document header that is valid for the whole document.
	ProcessKind_String	
10	[0..1] period.timeInterval	The start and end date and time for a given interval.
	ESMP_DateTimeInterval	--- The time interval that is associated with an electronic document and which is valid for the whole document.
11	[0..1] period.resolution	The number of units of time that compose an individual step within a period.
	Duration	--- The time interval that is associated with an electronic document and which is valid for the whole document.

##MXM##

!# shows all association ends of OpenADRRECMMOE_MarketDocument with other classes.

Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly
model::OpenADRRECMMOE_MarketDocument with other classes

Order	mult.	Class name / Role	Description
12	[0..*]	TimeSeries_Series	The time series that is associated with an electronic document. Association Based On : TimeSeries_Series CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::OpenADRRECMMOE_MarketDocument.[] --- CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::TimeSeries_Series.TimeSeries_Series[0..*]

##MXM##

Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Point

Order	mult.	Attribute name / Attribute type	Description
0	[0..1]	position Position_Integer	A sequential value representing the relative position within a given time interval.

##MXM##

!# shows all association ends of Point with other classes.

Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Point with other classes

Order	mult.	Class name / Role	Description
1	[0..*]	Quantity Quantity	The Quantity information associated with a given Point. Association Based On : CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Point. --- CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Quantity.Quantity[0..*]

##MXM##

Quantity

Description of quantities needed in the data exchange.

The type of the quantity is described either by the role of the association or the quantityType attribute.

The quality attribute provides the information about the quality of the quantity (measured, estimated, etc.).

!# shows all attributes of Quantity.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Quantity

Order	mult.	Attribute name / Attribute type	Description
-------	-------	---------------------------------	-------------

0	[1..1] quantity Decimal	The quantity value. The association role provides the information about what is expressed.
1	[0..1] quality Quality_String	The description of the quality of the quantity.
2	[0..1] type String	The description of the type of the quantity.
3	[1..1] quantityDefinition.mRID String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
4	[0..1] quantityDefinition.quantityType String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy.
5	[0..1] quantityDefinition.commodityFlowOccurence CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements, or expected future values, e.g. forecast, schedule, instruction.
6	[0..1] quantityDefinition.quantityQualifier String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value, maximum measured value, forecast expected value, forecast lower bound, forecast upper bound, lower operational limit, upper operational limit.
7	[1..1] quantityDefinition.unit.name MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20).
##MXM##		

Series

A set of similar physical or conceptual objects defined for the same period or point of time.

!# shows all association ends of Series with other classes.

Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Series with other classes

Order	mult.	Class name / Role	Description
0	[0..*]	Series_Period	The time interval and resolution for a period associated with a TimeSeries.
		Series_Period	Association Based On : CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Series.[]

##MXM##

Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Series_Period
Order mult. Attribute name / Attribute type Description

0	[1..1]	resolution	The definition of the number of units of time that compose an individual step within a period.
Duration			
1	[1..1]	timeInterval	The start and end time of the period.
ESMP_DateTimeInterval			

##MXM##

!# shows all association ends of Series_Period with other classes.

Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::Series_Period with other classes

Order	mult.	Class name / Role	Description
2	[1..*]	Po	The Point information associated with a given Series_Period.within a TimeSeries.
		int	Association Based On : CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Series_Period.[]
		Point	---

CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Point.Point[1..*]

##MXM##

TimeSeries_Series

A set of similar physical or conceptual objects defined for the same period or point of time.

!# shows all attributes of TimeSeries_Series.

Table !# - Attributes of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::TimeSeries_Series		
Order	Attribute name / Attribute type	Description
0	[0..1] mRID	A unique identification of the time series.
	ID_String	In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides a unique identification in the context of a business exchange such as time series identification, bid identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
1	[0..1] businessType	The identification of the nature of the time series.
	BusinessKind_String	
2	[0..1] curveType	The identification of the coded representation of the type of curve being described.
	CurveType_String	
3	[0..1] resourceTimeSeries.value1ScheduleType	Value1ScheduleType documents two key features for a specific resource operating mode: (a) the energy flow convention for a positive number as either production or consumption of service and (b) the sense of how the resource is planned to be used or was used as either a target or setpoint or as a relative change to a defined level.
	CommodityDirectionKind	
4	[0..1] flowDirection.direction	The coded identification of the direction of energy flow.
	DirectionKind_String	--- The flow direction associated with a TimeSeries.
5	[0..1] registeredResource.mRID	The unique identification of a resource.
	ResourceID_String	In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries.
6	[0..1] registeredResource.name	The name is any free human readable and possibly non unique text naming the object.
	String	--- The identification of a resource associated with a TimeSeries.
7	[0..1] registeredResource.description	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

String		--- The identification of a resource associated with a TimeSeries.	
8	[0..1] measurement_Unit.name	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20).	
	MeasurementUnitKind_String	--- The unit of measure associated with the quantities in a TimeSeries.	
##MXM##			
!# shows all association ends of TimeSeries_Series with other classes.			
Table !# - Association ends of CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model::TimeSeries_Series with other classes			
Order	mult.	Class name / Role Description	
9	[0..*]	Series	
		Series	Association Based On : CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::TimeSeries_Series.[] --- CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Series.Series[0..*]
##MXM##			

Datatypes

The list of datatypes used for the CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document assembly model is as follows:

- ESMP_DateTimeInterval compound
- BusinessKind_String datatype, codelist BusinessTypeList
- CurveType_String datatype, codelist CurveTypeList
- DirectionKind_String datatype, codelist DirectionTypeList
- ESMPVersion_String datatype
- ID_String datatype
- MeasurementUnitKind_String datatype, codelist UnitOfMeasureTypeList
- PartyID_String datatype, codelist CodingSchemeTypeList
- Position_Integer datatype
- ProcessKind_String datatype, codelist ProcessTypeList
- Quality_String datatype, codelist QualityTypeList
- ResourceID_String datatype, codelist CodingSchemeTypeList
- YMDHM_DateTime datatype

CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model

Overview of the model

!# shows the model.





Figure !# - CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model

IsBasedOn relationships from the European style market profile

!# shows the traceability dependency of the classes used in this package towards the upper level.

Table !# - IsBasedOn dependency

Name	Complete IsBasedOn Path
event	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::event
FlowDirection	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::FlowDirection
MarketParticipant	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::MarketParticipant
Measure_Unit	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Measure_Unit
OpenADRRECMMOE_MarketDocument	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::OpenADRRECMMOE_MarketDocument
Point	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Point
Process	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Process
Quantity	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Quantity
QuantityDefinition	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::QuantityDefinition
RegisteredResource	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::RegisteredResource
ResourceTimeSeries	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope

	Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::ResourceTimeSeries
Series	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Series
Series_Period	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Series_Period
Time_Period	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::Time_Period
TimeSeries_Series	CIMProfiles::MarketProfiles::RD_Projects::CEEDS::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document::CEEDS_OpenADR_ReferenceEnergyCurveMinMaxOperatingEnvelope Document contextual model::TimeSeries_Series

##MXM##